



Final Assignment (Major Project)



WHAT YOU NEED TO DO

You must:

- Select ONE project idea (or your own)
 - Demonstrate how it works
 - Submit a structured report
-



PROJECT OPTIONS (Choose One)



1. Password Strength Analyzer

Build a tool that:

- Checks password strength
- Detects weak patterns (123, name, DOB)
- Suggests stronger alternatives

👉 Bonus:

- Add entropy calculation
 - Detect common leaked passwords
-



2. Password Manager (Basic)

Build a simple password manager that:

- Stores credentials securely
- Uses encryption (basic level acceptable)
- Allows add / view / delete

👉 Bonus:

- Master password protection
 - CLI interface
-

3. Wordlist Generator Tool

Build a tool that:

- Generates custom wordlists based on input (name, DOB, interests)
- Mimics attacker behavior

👉 Example input:

Name: Rahul

DOB: 1998

Interest: bike

👉 Output:

rahul123

bike1998

rahul@98

👉 Bonus:

- Pattern combinations
 - Export as `.txt`
-

4. Phishing Link Detector

Build a tool that:

- Detects suspicious URLs

Checks:

- Fake domains
- Use of IP instead of domain
- Suspicious keywords

👉 Bonus:

- Highlight why link is dangerous
-

5. XSS Scanner Tool

Build a tool that:

- Tests URLs for XSS vulnerabilities

Input:

`http://example.com/search?q=test`

👉 Tool injects payloads like:

`<script>alert(1)</script>`

👉 Output:

- Vulnerable / Not vulnerable
-

6. OSINT Tool (Information Gathering)

Build a tool that:

- Takes input (email / username / domain)
- Finds publicly available information

Examples:

- Social links
- Username reuse
- Metadata

👉 Bonus:

- Combine multiple sources
-



7. Subdomain Finder

Build a tool that:

- Finds subdomains of a given domain

Example:

example.com → admin.example.com, dev.example.com



Bonus:

- Use wordlist-based discovery
-



8. Login Brute Force Simulator (Safe)

Build a tool that:

- Simulates brute-force attack on a test login system



Must include:

- Rate limiting awareness
 - Delay control
-



9. Log Analyzer (Security Focused)

Build a tool that:

- Reads server logs
- Detects suspicious activity

Examples:

- Multiple failed logins
 - Unusual IPs
-

10. File Integrity Checker

Build a tool that:

- Detects file changes using hashing

👉 Example:

- Compare hash before & after
-

11. Custom Vulnerable Lab (Advanced Option)

Create your own:

- Vulnerable web app

Include:

- At least 2 vulnerabilities (XSS, weak auth, etc.)

👉 This is the **highest scoring option**

12. Your Own Idea (Approved)

You can propose your own tool if:

- It solves a cybersecurity problem
 - You explain it clearly
-



DELIVERABLES

1. Working Tool

- Code (Python / JS / PHP etc.)
 - Must run properly
-

2. Demonstration

- Screenshots OR screen recording
 - Show tool in action
-

3. Report (IMPORTANT)

Include:

♦ Problem Statement

What problem are you solving?

♦ How Your Tool Works

Explain logic and flow

♦ Features

List features implemented

♦ Output / Results

Show example usage

♦ Limitations

What your tool cannot do

♦ Future Improvements

What you would add next

INDIVIDUAL OR GROUP WORK

- You can work:
 - Individually
 - OR in a group (max 3 members recommended)
-

If Working in a Group

You **MUST** include in your PPT:

- Names of all members
- Role of each member
- Contribution of each member

👉 This is **mandatory**

EVALUATION CRITERIA

Criteria	Weight
Functionality	30%
Problem-solving	20%
Creativity	20%
Report quality	15%
UI / UX / Usability	15%



RULES

- No plagiarism
- No unsafe or illegal testing
- Keep everything ethical and controlled



OPTION 2: Research-Based Case Study (Non-Coding Option)

For students who prefer analysis over development.



What You Need to Do

Choose a **real-world cybersecurity topic, attack, or incident** and create a detailed case study.



Suggested Topics



Real-World Cyber Attacks



Phishing & Social Engineering



Password Attacks



Web Vulnerabilities



OSINT & Digital Footprint



AI in Cybersecurity



Data Breaches



Mobile Security



Cloud Security



Case Study Must Include

- What is the topic?
- How the attack/issue works
- Real-world example
- Impact (technical + business)
- Prevention & mitigation
- Your analysis (important)



DELIVERABLES

1. PowerPoint Presentation (MANDATORY)

Must include:

- Problem / Topic
 - Approach / Explanation
 - Demo (if tool) OR Case Study (if research)
 - Key findings
-

2. Project Output

- Detailed case study content
-



EVALUATION CRITERIA

Criteria	Weight
Understanding / Depth	25%
Execution (Tool or Research)	25%
Creativity / Insights	20%
Presentation (PPT)	20%
Clarity & Communication	10%



RULES

- No plagiarism
 - No unsafe or illegal testing
 - Keep everything ethical and controlled
-



PRO TIPS

- Choose something you understand
- Don't overcomplicate
- Focus on **clarity + execution**

Submission Link: [CLICK ME](#)

Or

<https://forms.gle/QmhMpznxL7fqNeAm8>